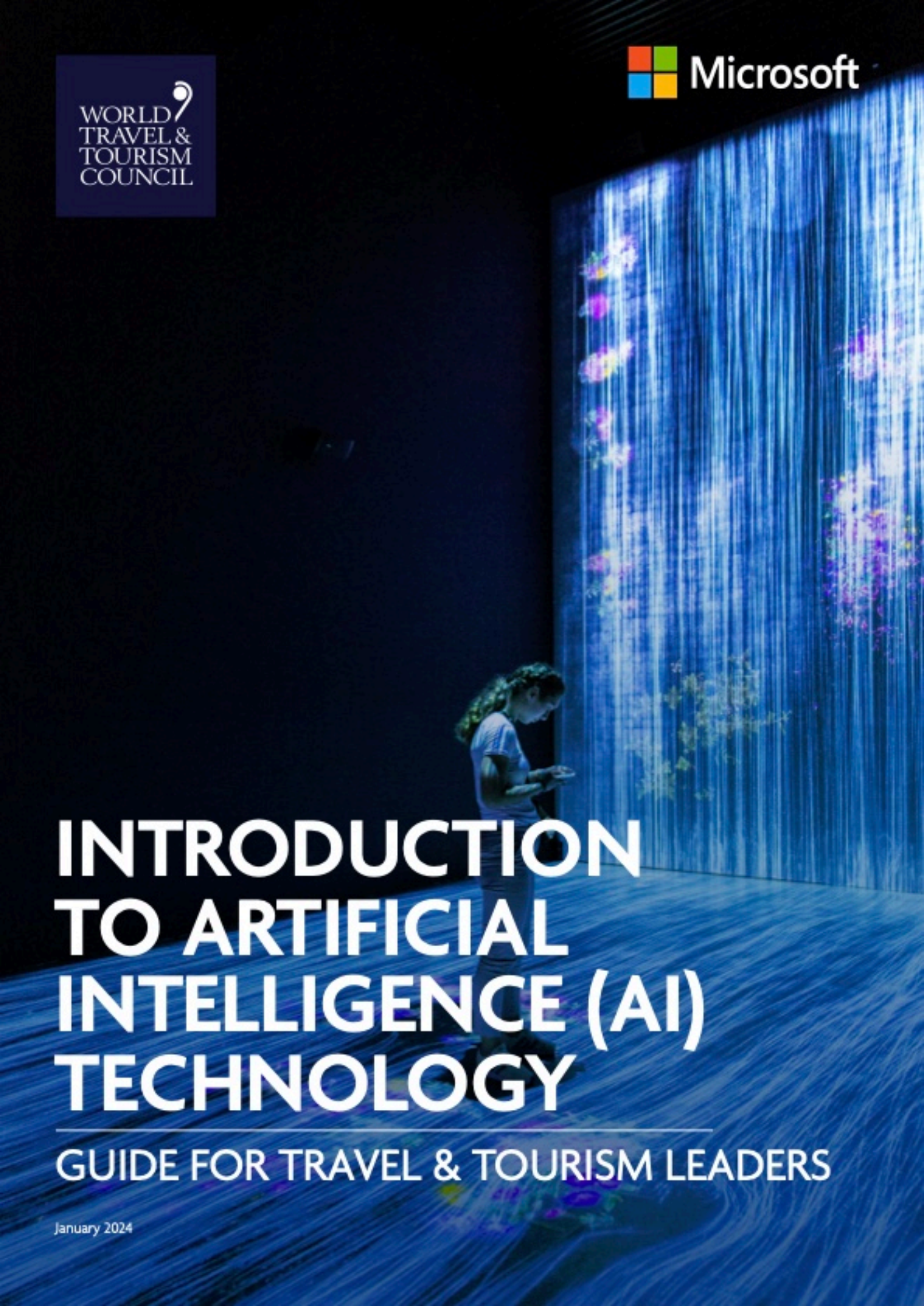
WORLD TRAVEL & TOURISM COUNCIL

Microsoft



INTRODUCTION TO ARTIFICIAL INTELLIGENCE (AI) TECHNOLOGY

GUIDE FOR TRAVEL & TOURISM LEADERS

January 2024



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FOREWORD

In 1950, codebreaker and now celebrated computer scientist, Alan Turing, posed one of the most interesting questions ever faced by humankind - can machines think?

Artificial Intelligence (AI) has occupied scientists for decades. But in 2023, it caught the attention of global businesses, policymakers and civil society alike. In the UK, King Charles III called it "one of the greatest technological leaps in the history of human endeavour". The UN Secretary General described AI as "unprecedented" and even the Pope stated that AI "has the potential to contribute in a positive way to the future of humanity" before dedicating World Peace Day 2023 to the use of AI.

As business leaders in Travel & Tourism it is vital that we understand the opportunities and challenges of this transformative technology. I am therefore delighted that the World Travel & Tourism Council (WTTC) has partnered with Microsoft, one of our members and a leading technology company, to bring you the very latest insights into AI and what it could mean for our sector.

What follows is the first report in a four-part initial series on artificial intelligence. We begin with an introduction to AI: how it works, where it came from and the different types of AI available today. Three further publications will explore how AI is currently being used in Travel & Tourism, and the fast-moving world of AI governance, safety and regulation.

By understanding these trends, we aim to help Travel & Tourism business leaders to prepare for the future and navigate the complexities of AI. But this report isn't just technical facts and figures. We also examine how AI can help tourism grow and do good over the years to come, opening up new and exciting possibilities for personalised, seamless and sustainable travel experiences.

Ultimately, AI is just an engine. The task now is to ensure that it fuels us towards a destination of our choosing – a sector that is greener, safer, more prosperous and more free. After all, as Alan Turing said back in 1949, we may be experiencing "only a foretaste of what is to come, and only the shadow of what is going to be". If we get it right with AI, that destination could be very bright indeed.



Julia Simpson
President & CEO
World Travel & Tourism Council



INTRODUCTION

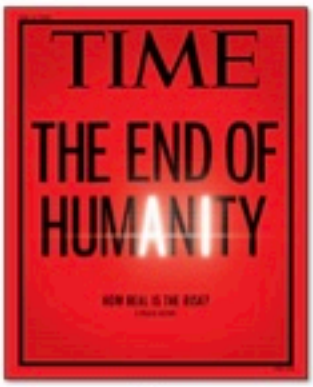
"...It happened gradually, then suddenly..."

This famous quote from Ernest Hemingway's 1926 novel, *The Sun Also Rises* could be used to describe some of the world's most profound technological changes. Small advancements accumulate and then all of a sudden, the world is a different place.

That could describe the world before and after the birth of the internet and could now be applied to Artificial Intelligence (AI), which has 'gradually, then suddenly' burst onto the scene after nearly a century of research. It is almost impossible to browse the news or social media today, without seeing mention of Artificial Intelligence and the magazines Time, Science, Cosmopolitan and The Economist (to name only a few) have all dedicated cover stories to AI in 2023.



Cosmopolitan
www.cosmopolitan.com



Time
https://time.com



Economist
www.economist.com



Science
www.science.org

But where has AI come from and why is it suddenly such big news?

This report aims to answer those questions, by providing a general introduction to Artificial Intelligence technology, its business applications and its relevance to Travel & Tourism.

BRIEF HISTORY OF ARTIFICIAL INTELLIGENCE (AI)

AI has gained significant attention in recent years – and especially in 2023 – but AI is not new and can trace its history back to the development of computers after the Second World War, with the Dartmouth Conference in 1956 bringing together researchers from multiple fields to explore "thinking machines". This is widely considered the start of AI as a distinct field of study and where the term "Artificial Intelligence" was used for the first time by the visionaries at that conference.

But it was not until the turn of the century that AI really came to the public's attention, when the **IBM Deep Blue** supercomputer beat chess grandmaster Gary Kasparov in 1997, with artificial intelligence algorithms developed by IBM engineers. A few years later, in 2011, the **IBM Watson** computer won the US gameshow Jeopardy, after

being trained on a huge data set. These events showed the world for the very first time that computers powered with large amounts of data and artificial intelligence software – albeit very powerful computers out of reach of the general public - were capable of outperforming humans at very complex tasks.

Over the past decade several companies have taken this a step further and developed AI systems that have achieved incredible results and performed tasks not possible by humans, due to the huge scale and complexity of the challenge. **AlphaFold** from **Google Deepmind** is one example. The programme uses AI to predict the 3D structure of nearly every protein in the human body. This improves our understanding of diseases like Alzheimer's and can massively accelerate research into medicines, vaccines and drugs. Healthcare is therefore one of the most important and useful applications of AI.

In 2018 a painting called "Edmond de Belamy, from La Famille de Belamy" was sold at Christie's auction house in New York for a staggering $432,500 (USD). It was generated with artificial intelligence after being trained on many images of portraits from the 18th and 19th centuries and was the first piece of AI art sold at auction. The sale sparked a significant debate about the future role of humans in creative professions, such as art and digital media.



AI Painting (2018)
Edmond de Belamy



AI Painting (2023)
Girl with Glowing Earrings



AI Photo (2023)
Pseudomnesia : The Electrician

But this was only the start of a 'boom' in AI generated creative content. In 2023, a museum in The Hague (Netherlands) loaned the world famous, *Girl with Pearl Earring* (c. 1665) by Johannes Vermeer to an international exhibition and temporarily replaced it in their gallery with an AI inspired version entitled *Girl with Glowing Earrings*, with many visitors to the gallery believing it was a real painting. In April 2023 the German artist Boris Eldagsen won the 2023 Sony World Photography Award with an image entitled *Pseudomnesia: The Electrician*, before refusing the prize after revealing it was created by AI. (Pseudomnesia is Latin for fake memory!)

These images show how far AI creative technology has advanced in only a few short years, from an ability to generate low quality AI portraits in 2018, to high quality AI paintings and photorealistic AI images today. Both the *Girl with Pearl Earring* and *Pseudomnesia* images were created with AI software that is now available to everyone online and is partly why AI has taken off with the public in 2023, enabling any of us to become potential artists.

But paintings and photography are not the only form of art that has been shaped by AI. In 2023 the Grammys announced that music created with AI would be eligible for awards and Paul McCartney announced he had used AI to extract John Lennon's voice from an old demo tape. In late 2023, this track was released as the final Beatles Song *Now and Then* [1] which drew considerable public attention to AI across all generations. Canadian singer Grimes has also invited anyone to create new songs with an AI copy of her voice, offering to split the royalties equally on any AI track that is commercially successful.

Going a step further, an area of AI that is still in development is AI generated video. Imagine being able to produce a corporate marketing video easily and in only a few moments on your office laptop, or even turning your favourite book into a movie using AI. Whilst AI generated video does not currently exist at this advanced level yet, it is clear that AI is rapidly moving forward and has huge potential for both businesses and society.

But there are also risks and concerns about the widescale adoption of AI, which are covered later in this report. The use of AI has led to some highly publicised events, such as a strike by Hollywood actors and writers in 2023, partly due to concerns about AI's impact on the movie industry. Their concerns include the ability of AI to write scripts and the potential to create an AI digital replica of an actor (or extra), which could be included in films, without the physical actor present and potentially long after they have died. The full impact of AI on jobs of the future is yet to be determined, but it is possible that labour strikes will be experienced in other industries as AI rolls out further into society and other areas of work.

There is also a concern that AI could be used for disinformation (the deliberate sharing of false information), especially to interfere with political elections. High quality, photorealistic – but fake – images have already gone viral on the internet, such as an AI generated image of the Pope wearing a puffer jacket and an image purporting to show the arrest of former US President Donald Trump. Both images were generated by AI and both are fake, but they are photorealistic, very convincing, generated on a home computer in only a few minutes and were shared very widely across the internet and social media. It is therefore clear that the wide scale implementation of AI across society needs 'guardrails' for its appropriate use, which could range from formal regulations to codes of conduct and best practice guidance. These options are discussed further in an accompanying WTTC report on '**AI Risks & Governance**'.



Viral AI-generated (fake) images of the Pope and former US President Donald Trump

But not all celebrities are against AI and digital replicas – often known as 'deepfakes'. For example World Cup winning Argentinian footballer Lionel Messi has allowed a deepfake version of himself to advertise snacks, as part of his sponsorship deal with PepsiCo. At the website www.messimessages.com anyone can generate and download to their phone a short personalised deepfake video message from Messi who invites your friends by name, to bring food and watch the big game with him – and it can even get Messi to provide the message in his voice in English, Spanish, Portuguese or Turkish!



Argentinian footballer Lionel Messi : Deepfake Messi Messages

AI has therefore been regularly hitting the news headlines and as with all new trends, a few high profile cases can propel a technology that is initially only used by a few early adopters, into global mass adoption. For AI this happened with the international release of a sophisticated AI powered chatbot, named **ChatGPT**, in November 2022 from a company called OpenAI. This caught the public's attention, registering over 100 million users in only 2 months and making it (at the time) the fastest growing consumer product of all time. This tipped AI into the public mainstream in 2023 and sparked the current worldwide interest in AI.

While ChatGPT was the first AI chatbot to reach mass consumption, others are now available including **Google Bard**, **IBM Watsonx.ai** and **Microsoft Copilot**, with others produced around the world, such the **Ernie** AI chatbot from Baidu in China.

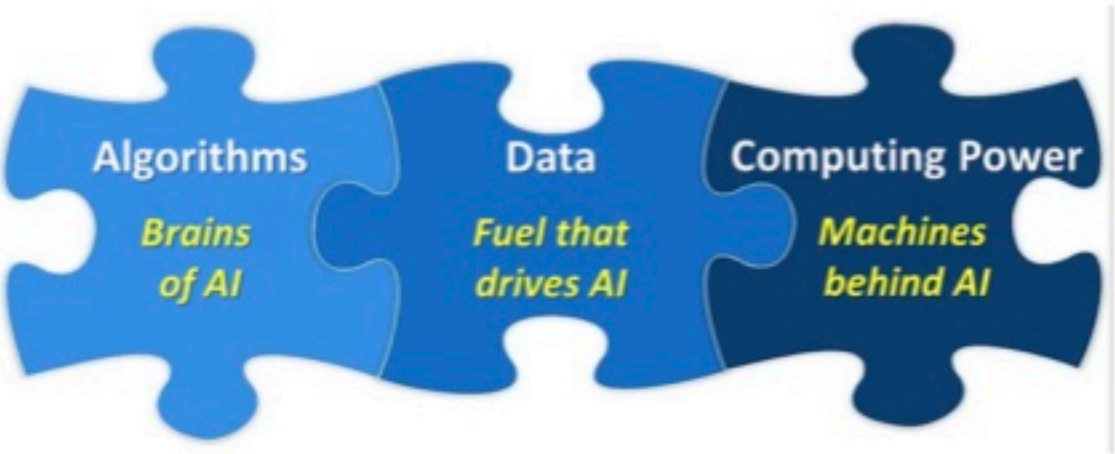
These sophisticated AI chatbots are not the only way to use AI, but they have exposed the power of AI to the general public, which everyone can now access on their desktop computers and smartphones (and through digital connections to the 'cloud'), unlike the 1990's when the use of AI was limited to a very small number of owners of supercomputers (such as IBM Deep Blue). The future of AI is exciting and evolving very fast. It is impossible to accurately predict how AI will shape our lives over the next 5 to 50 years, but the following section explains why now is a crucial moment for the future of AI and society.

WHAT IS ARTIFICIAL INTELLIGENCE & WHY THERE IS GLOBAL INTEREST NOW

One of the most important aspects of **AI is that it is a multi-use technology**. Like electricity it can be applied in lots of different ways, to lots of different scenarios.

There is no single, universally accepted definition for Artificial Intelligence, but the Oxford English Dictionary defines AI as *"the capacity of computers, or other machines, to exhibit intelligent behaviour"*. This means AI systems appear to think, learn and act like humans and in some cases exceed the capabilities of humans. AI systems can analyse vast amounts of data, solve complex problems, make decisions and perform creative tasks.

Some AI technologies have been around for more than 50 years, but advances in computing power, the availability of enormous quantities of data and new developments in software algorithms have led to major AI breakthroughs in recent years. It is these three components of **advanced algorithms**, **data** and **computing power**, that explain how machines can exhibit intelligent behaviour and why AI has suddenly exploded into our everyday lives.



Algorithms tell computers what to do. Data tells computers what to learn. Computing power gives machines the power to learn and make decisions

ALGORITHMS : THE BRAINS OF AI

'Traditional programming' involves encoding human knowledge and experience into a set of precise rules that a computer can follow, step-by-step, which make the computer appear to respond intelligently.

These **rules, called algorithms, tell computers how to perform tasks** and in traditional programming are often expressed in an 'IF-THEN-ELSE' format, which resembles a decision tree. For example to create a 'digital doctor', an algorithm might look like:

```
IF the patient has a fever
    THEN prescribe Drug X
ELSE send the patient home
```

The intelligence in traditional computer systems comes directly from human knowledge and expertise being recorded into a format that a computer can process. Humans can read this 'computer code' and understand how these digital systems make specific decisions and can therefore identify mistakes and improve the system. For example, in the 'digital doctor' scenario, new clauses could be added to deal with special cases, or to reflect new medical knowledge. For example:

```
IF the patient has a fever AND is allergic to Drug X
    THEN prescribe Drug Y
ELSE prescribe Drug X
```